TCP vs UDP Comparison

1. Introduction

TCP and UDP are transport-layer protocols used to transfer data between applications. This project implements both protocols for communication between a Vehicle ECU and Dashboard ECU using Linux socket programming.

2. TCP Communication

TCP is connection-oriented. A connection is established before application data is transmitted.

In this project: - The Dashboard ECU acts as the TCP server. - The Vehicle ECU acts as the TCP client. - The server listens on port 5000. - Vehicle data is transmitted using the send() function. - The Dashboard receives data using recv(). - TCP provides reliable, ordered byte-stream delivery.

3. UDP Communication

UDP is connectionless and sends individual datagrams.

In this project: - The Dashboard ECU receives UDP datagrams on port 5000. - The Vehicle ECU transmits data using sendto(). - The Dashboard receives data using recvfrom(). - UDP has lower protocol overhead than TCP but does not guarantee delivery or ordering.

4. Comparison

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable delivery mechanisms	No delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Data format	Byte stream	Datagrams
Connection setup	Required	Not required
Typical socket type	SOCK_STREAM	SOCK_DGRAM
Send function	send()	sendto()
Receive function	recv()	recvfrom()
Server setup	bind(), listen(), accept()	bind()
Project use	Reliable vehicle data transfer	Lightweight data transfer

5. Wireshark Analysis

The UDP capture shows packets sent from 127.0.0.1 to 127.0.0.1 with destination port 5000.

The TCP capture shows an established TCP connection between client port 37170 and server port 5000. TCP data packets contain the vehicle speed, RPM, and steering-angle values. Acknowledgment packets are also visible.

6. Conclusion

Both TCP and UDP communication were implemented using Linux sockets. Wireshark was used to inspect the transmitted vehicle-monitoring data. The experiment demonstrates the differences between connection-oriented TCP communication and connectionless UDP communication.

The tests were performed over the loopback interface. Therefore, the results demonstrate local socket communication rather than testing a physical Ethernet network.